

Rule & Context Tables – Authoring Guidelines (Thesis Version)

This document provides **clear, operational guidance** for constructing the two core artifacts used in this thesis:

1. **Rule Table (Enforceable Claims)** – what the system *claims it can verify*
2. **Context Table (Interpretative Sources)** – what the rules *mean, rely on, or are constrained by*

The separation is intentional: **not all legal text is enforceable, and not all enforceable claims should restate the law.**

1. Conceptual Boundary (Read This First)

Rule Table

- Contains **only enforceable claims**.
- Every row represents something that can be **checked, tested, or evidenced**. (by the financial institutions)
- Every rule must be defensible in an audit discussion.

If you are not comfortable answering “*How do we verify this?*”, it does **not** belong in the Rule Table.

Context Table

- Contains **interpretative material**.
- Context entries **do not produce pass/fail outcomes**.
- They exist to explain **definitions, intent, scope, rationale, and consequences**.

Context supports rules; it never replaces them.

2. Rule Table – Authoring Guidelines

Enforceable vs. Non-Enforceable: Concrete Examples

The distinction between *enforceable* and *non-enforceable* content is best understood through examples. The table below illustrates how similar regulatory language is treated differently depending on whether it can be verified or evidenced.

Examples of Enforceable Rules (Belong in Rule Table)

Regulatory intent	Why it is enforceable	Example Rule Definition
Prohibit an action	A single occurrence can be conclusively detected	Rule: Transfers to unverified addresses must not be allowed. obligation_nature: prohibition check_method: deterministic pass_criteria: binary
Require a control	Existence or configuration can be checked	Rule: An identity registry must exist and be referenced during transfers. obligation_nature: requirement check_method: deterministic pass_criteria: binary
Require a compliance capability	Presence of auditable signals can be observed	Rule: The system must be able to support ongoing transaction monitoring. obligation_nature: capability check_method: observational pass_criteria: evidence-based

Examples of Non-Enforceable Content (Belong in Context Table)

Regulatory text	Why it is not enforceable	How it is captured
"Institutions should adopt a risk-based approach..."	Depends on judgement and adequacy	Context (GUIDANCE)
"Monitoring should be appropriate having regard to risk"	No objective pass/fail condition	Context (OBLIGATION_TEXT or GUIDANCE)
"Failure to comply constitutes an offence..."	Describes consequences, not behaviour	Context (CONSEQUENCE)
"In this Ordinance, 'customer' means..."	Defines terms only	Context (DEFINITION)

Practical Rule of Thumb

If you can fill in **all three** of the following with confidence, the item belongs in the Rule Table:

1. *What exactly is being claimed?*
2. *How is it checked?*
3. *What evidence is sufficient to say it passes?*

If any answer is unclear, the content belongs in the Context Table.

Purpose

The Rule Table defines **enforceable compliance claims** derived from regulatory text and system design assumptions.

Each rule answers: > *What can we legitimately verify about compliance behaviour or capability?*

Purpose

The Rule Table defines **enforceable compliance claims** derived from regulatory text and system design assumptions.

Each rule answers: > *What can we legitimately verify about compliance behaviour or capability?*

Rule Table Schema

Field	Required	Description	Original column name
rule_id	Yes	Stable unique identifier (machine-friendly)	N/A
rule_ref	Yes	Reference to the originating regulatory section, schedule, or clause title (e.g. Division 2 – CDD Obligation, Schedule 2 §6 Ongoing Monitoring)	Provision or schedule
compliance_category	Yes	High-level compliance domain the rule belongs to	N/A
rule_objective	Yes	High-level purpose of the rule	Compliance objective
original_text	Yes	Key regulatory text fragment that motivated the rule	N/A

obligation_nature	Yes	See explanation below	Must have / must not have
check_method	Yes	See explanation below	N/A
pass_criteria	Yes	See explanation below	N/A
enforcement_interface	Optional	Where / how enforcement occurs (e.g. IdentityRegistry + TrustedIssuersRegistry + ClaimTopicsRegistry)	ERC-3643 (suggested modules / functions)

Rule Objective – How to Write

rule_objective captures the **compliance objective (policy intent)** of the rule. It should be written in **regulatory / compliance language** and should not describe implementation details.

Examples: - “Restrict token ownership and transfer to eligible, verified participants.” - “Ensure administrative controls exist to freeze or enforce restrictions when required.”

Implementation details belong in enforcement_interface.

Enforcement Interface – How to Write

enforcement_interface describes **where and how enforcement is implemented** (e.g., ERC-3643 component, hook, module, function/event interface, or operational gateway).

Guidelines: - Be concrete and system-oriented (interfaces, modules, hooks, roles). - Avoid restating the compliance intent. - If enforcement is hybrid, describe both on-chain and off-chain touchpoints.

Examples: - “ERC-3643 Compliance module: transfer hook veto based on IdentityRegistry verification status.” - “Admin role can freeze addresses via agent function; actions logged via events.”

Obligation Nature – How to Choose

Use **exactly one** of the following:

1. prohibition

- Something **must never occur**.
- Typical phrasing: *must not, shall not, no person may*.

- Usually paired with **deterministic + binary** checks.

Example: > Transfers to unverified addresses must not be allowed.

2. requirement

- A control or condition **must exist**.
- Focuses on presence or configuration, not behaviour quality.

Example: > An identity registry must exist and be referenced during transfers.

3. capability

- The system must be **able to support or demonstrate** a compliance function.
- Does **not** claim effectiveness or outcomes.

Example: > The system must be able to support ongoing monitoring through observable signals.

Check Method

Question it answers: How is this rule verified during audit?

Value	Meaning
deterministic	System behaviour can be conclusively evaluated automatically
observational	Only observable evidence can be assessed

Allowed values: - **deterministic** – Compliance can be conclusively checked by system behaviour or configuration. - **observational** – Compliance is assessed via observable evidence or signals rather than a single definitive outcome.

Examples: - *deterministic*: A transfer transaction reverts when the recipient is unverified.
- *observational*: Events or hooks indicate that monitoring logic is executed.

Pass Criteria

Question it answers: What form of proof is sufficient to say this rule is satisfied?

Value	Meaning
binary	Pass/Fail can be directly concluded
evidence-based	Presence of auditable evidence determines satisfaction

Allowed values: - **binary** – The rule can be directly marked PASS or FAIL. - **evidence-based** – Compliance is supported by the presence of auditable evidence, not a final judgement.

Examples: - *binary*: Transfer either succeeds (FAIL) or reverts (PASS). - *evidence-based*: Monitoring events or logs are present for inspection.

Important: - capability rules are almost always evidence-based. - Do **not** use binary unless you are confident a definitive conclusion is justified.

Compliance Category

Question it answers: Which compliance domain does this rule belong to?

Allowed values:

- **IDENTITY_ELIGIBILITY** – Who is allowed to hold or receive tokens.
- **TRUST_ANCHOR** – Identity issuers, claims, registries, or trust roots.
- **TRANSFER_COMPLIANCE** – Conditions governing token transfers.
- **JURISDICTION_RESTRICTION** – Geographic or residency-based restrictions.
- **TRANSACTION_LIMITS** – Amount, frequency, or exposure limits.
- **ADMIN_ENFORCEMENT** – Administrative or emergency controls.

This field is for **thematic grouping**, not legal interpretation.

Examples: - *IDENTITY_ELIGIBILITY*: Only verified investors may receive tokens. - *TRANSFER_COMPLIANCE*: Transfers must be blocked if compliance checks fail.

Rule Authoring Checklist

Before adding a rule, confirm: - [] Can this be verified or evidenced? - [] Am I avoiding claims about effectiveness or adequacy? - [] Is the obligation framed as prohibition, requirement, or capability? - [] Would I be comfortable defending this rule to an auditor?

If any answer is “no”, move the content to the Context Table instead.

3. Context Table – Authoring Guidelines

Purpose

The Context Table stores **interpretative sources** that explain, justify, or constrain rules.

Context entries answer: > *Why does this rule exist, and how should it be understood?*

Context Table Schema

Field	Required	Description
context_id	Yes	Stable unique identifier
source_ref	Yes	Precise legal reference (Division / Schedule / Section)
context_type	Yes	See enum below
text	Yes	Source text or faithful paraphrase
notes	Optional	Interpretation, rationale, or exclusions

Context Type – How to Classify

Use **one** of the following:

DEFINITION

- Statutory definitions or interpretation clauses.
- Never converted directly into rules.

OBLIGATION_TEXT

- Legal obligations that are **not directly enforceable as system claims**.
- Often a source from which rules are later abstracted.

GUIDANCE

- Risk-based, discretionary, or “should”-style language.

- Supports interpretation but does not create enforceable claims.

CONSEQUENCE

- Penalties, offences, or liability statements.
- Used to justify severity, never to define enforcement logic.

SCOPE_NOTE

- Explicit exclusions, regulator powers, or out-of-scope clauses.
- Prevents misinterpretation and reviewer confusion.

MAPPING_NOTE

- Explanation of how specific text was transformed into a rule.
 - Strongly recommended for transparency and defence.
-

When Content Belongs Only in Context

Use the Context Table when: - The text is descriptive rather than enforceable - The obligation depends on human judgement or risk assessment - The clause explains *how* or *why*, not *what the system must enforce*

4. Relationship Between the Tables

- Rules **may reference** Context entries conceptually.
- Context entries **do not depend** on rules.
- Traceability is achieved through rule_ref, source_ref, and narrative alignment, not bidirectional keys.

In this thesis version, the separation is logical rather than relational.

5. Guiding Principle (Final)

Rules state what can be verified. Context explains what must be understood.

Maintaining this boundary is essential for methodological clarity, audit defensibility, and future system extensibility.